

Skills Summary

Computing Distance and Midpoint from Coordinates

Use this reference while completing your worksheet or when studying.

Skill 1 — The Distance Formula

Formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Why: the horizontal leg is $|x_2 - x_1|$, the vertical leg is $|y_2 - y_1|$, and the segment is the hypotenuse — so this *is* the Pythagorean theorem written in coordinates.

- Order does not matter: squaring removes the sign, so A to B equals B to A .
- Leave $\sqrt{\quad}$ exact unless a decimal is requested; simplify with the largest perfect-square factor.

Example: Find the distance from $(2, 1)$ to $(10, 7)$.

Step 1. Two points and a length wanted, so build the right triangle: subtract the x 's for one leg and the y 's for the other.

Step 2. Legs $10 - 2 = 8$ and $7 - 1 = 6$, so $d = \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = \sqrt{100} = 10$

Try it: Find the distance from $(-1, 3)$ to $(3, 6)$.

check: 5

Skill 2 — The Midpoint Formula

Formula: $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

Why: a midpoint is not a length — it is the *average* of the endpoints, one coordinate at a time. So you **add** here, where the distance formula subtracts.

- The answer is a **point** (x, y) , never a single number.
- Each coordinate of M must land between the two given ones — a fast sanity check.

Example: Find the midpoint of $(-4, 7)$ and $(6, 1)$.

Step 1. The point halfway along a segment is the average of the endpoints, so average the x 's and the y 's separately.

Step 2. $\frac{-4 + 6}{2} = 1$ and $\frac{7 + 1}{2} = 4$, and each lands between its endpoints, so $M = (1, 4)$

Try it: Find the midpoint of $(3, -8)$ and $(9, 2)$.

check: $(\frac{3+9}{2}, \frac{-8+2}{2}) = M$

Skill 3 — Using Midpoints in a Coordinate Proof

The move: to prove two segments **bisect each other**, compute the midpoint of each. If the two midpoints are the *same point*, that point lies on both segments and halves both, which is exactly what bisecting means.

Related coordinate claims: equal distances prove congruent sides (isosceles); a midpoint plus a distance gives the length of a median.

Example: Parallelogram $A(0,0)$, $B(8,2)$, $C(10,6)$, $D(2,4)$: do the diagonals bisect each other?

Step 1. Bisecting each other means the two diagonals share one midpoint, so compute the midpoint of each diagonal and compare.

Step 2. Diagonal $\overline{AC}$: $\left(\frac{0+10}{2}, \frac{0+6}{2}\right) = (5, 3)$, and diagonal $\overline{BD}$ gives the same point, so they bisect each other.

Try it: One diagonal runs from $(-2,1)$ to $(6,9)$ and the other from $(0,7)$ to $(4,3)$. Find the shared midpoint.

(5, 2) :rcəp

(!) Watch out: Do not add the legs. $\sqrt{5^2 + 12^2} = 13$, not $5 + 12 = 17$. And keep the two formulas apart: distance *subtracts* then squares, midpoint *adds* then halves.